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Homework 1

1. Let A = {x, y, z} and B = {x, y}.

a. no. A has the element z, but z is not in B.

b. yes. every element in B is also in A.

c. $A \cup B = \{x, y, z\}$ since the union includes all elements from both sets without repeating anything.

d. $A \cap B = \{x, y\}$ because these are the elements that appear in both sets.

e. $A \times B = \{(x,x), (x,y), (y,x), (y,y), (z,x), (z,y)\}$. this comes from pairing each element of A with every element of B.

f. the power set of B is $\{\emptyset, \{x\}, \{y\}, \{x, y\}\}$ since these are all the possible subsets of B.

2. If set A has a elements and set B has b elements, how many elements are in $A \times B$?

$A \times B$ has $a \cdot b$ elements. for every element in A, there are b possible elements in B to pair with, so the total number of ordered pairs ends up being $a \cdot b$.

3. If set C has c elements, how many elements are in the power set of C?

the power set has 2^c elements. each element in C has two choices when making a subset, either included or not included, so the total number of subsets is 2^c.

4. Find the error in the proof that 2 = 1.

the error happens when dividing by $(a – b)$. since the proof starts by assuming $a = b$, that means $(a – b) = 0$, and dividing by zero is not allowed.

5. Let $S(n) = 1 + 2 + … + n$ and $C(n) = 1^3 + 2^3 + … + n^3$.

a. prove that $S(n) = n(n + 1)/2$.

first check the base case $n = 1$. $S(1) = 1$ and plugging into the formula gives $1(2)/2 = 1$, so it works. now assume that for some $k \geq 1$, $S(k) = k(k + 1)/2$. for the next step, $S(k + 1) = S(k) + (k + 1)$. substituting the assumption gives $k(k + 1)/2 + (k + 1)$. factoring out $(k + 1)$ results in $(k + 1)(k/2 + 1)$, which simplifies to $(k + 1)(k + 2)/2$. this matches the formula when $n = k + 1$, so the statement holds for all natural numbers by induction.

b. prove that $C(n) = (n(n + 1)/2)^2$.

start with the base case $n = 1$. $C(1) = 1^3 = 1$ and the formula gives $(1(2)/2)^2 = 1$, so the base case works. assume for some $k \geq 1$ that $C(k) = (k(k + 1)/2)^2$. then $C(k + 1) = C(k) + (k + 1)^3$. substituting the assumption gives $(k(k + 1)/2)^2 + (k + 1)^3$. factoring out $(k + 1)^2$ leaves $(k + 1)^2[k^2/4 + (k + 1)]$. combining terms inside the brackets gives $(k + 1)^2[(k^2 + 4k + 4)/4]$, which simplifies to $((k + 1)(k + 2)/2)^2$. this matches the formula when $n = k + 1$, so by induction the statement holds for all natural numbers.